HDL-cholesterol-raising effect of orange juice in subjects with hypercholesterolemia.

BACKGROUND: Orange juice-a rich source of vitamin C, folate, and flavonoids such as hesperidin-induces hypocholesterolemic responses in animals. OBJECTIVE: We determined whether orange juice beneficially altered blood lipids in subjects with moderate hypercholesterolemia. DESIGN: The sample consisted of 16 healthy men and 9 healthy women with elevated plasma total and LDL-cholesterol and normal plasma triacylglycerol concentrations. Participants incorporated 1, 2, or 3 cups (250 mL each) of orange juice sequentially into their diets, each dose over a period of 4 wk. This was followed by a 5-wk washout period. Plasma lipid, folate, homocyst(e)ine, and vitamin C (a compliance marker) concentrations were measured at baseline, after each treatment, and after the washout period. RESULTS: Consumption of 750 mL but not of 250 or 500 mL orange juice daily increased HDL-cholesterol concentrations by 21% (P: < 0.001), triacylglycerol concentrations by 30% (from 1.56 +/- 0.72 to 2.03 +/- 0.91 mmol/L; P: < 0.02), and folate concentrations by 18% (P: < 0.01); decreased the LDL-HDL cholesterol ratio by 16% (P: < 0.005); and did not affect homocyst(e)ine concentrations. Plasma vitamin C concentrations increased significantly during each dietary period (2.1, 3.1, and 3.8 times, respectively). CONCLUSIONS: Orange juice (750 mL/d) improved blood lipid profiles in hypercholesterolemic subjects, confirming recommendations to consume >/=5-10 servings of fruit and vegetables daily.